

Abdul-Kazeem Shamba

(+47) 925 59 431

Trondheim, Norway

abdul.k.shamba@ntnu.no

Robotic Engineer
Artificial Intelligence Researcher

GitHub: ShambaI
LinkedIn: abdul-kazeem-shamba

Doctoral Researcher in Computer Science at NTNU with expertise in representation learning, computer vision, and multimodal sensor data for robotics applications. Holds four advanced degrees across AI, robotics, and engineering, including an MSc in Marine Intelligent Robotics and ECE from Carnegie Mellon. Experienced in developing full training pipelines, deploying real-world robotics and vision systems, and publishing in top AI conferences. Strong background in self-supervised and contrastive learning, satellite data visualization, and autonomous systems, with hands-on skills in Python, PyTorch, ROS, C++, Docker, and embedded control.

EDUCATION

Doctor of Philosophy (PhD) in Computer Science , Norwegian University of Science & Tech., Norway	2023 — Present
Master of Science in Marine Intelligent Robotics , Norwegian University of Science & Tech., Norway	2022 — 2023
Master of Science in Marine and Maritime Intelligent Robotics , Univeristy of Toulon, France	2021 — 2022
Master of Science in Electrical and Computer Engineering , Carnegie Mellon University, USA	2020 — 2021
Bachelor of Science in Electronic and Electrical Engineering , Obafemi Awolowo University, Nigeria	2013 — 2018

- **PhD's Tentative Thesis:** "Representation Learning for Predictive Analytics and Reasoning in Multimodal Sensor Data"
- **Master's Thesis:** "Generating Realistic Underwater Renderings using Generative Adversarial Networks" - Grade A
- **Bachelor's Thesis:** "Design and Development of an Arm Prosthesis using Electromyogram Control" - Grade A
- **Standardize Test Scores:** GRE 317 (Q162 V155 AWA4) & TOEFL 101 (R27 L27 S22 W25)

Top Courses: Intelligent Agents, Probabilistic Artificial Intelligence, Visual Computing (3D Graphics Rendering), Autonomous Marine Systems, Introduction to Visual SLAM, Artificial Intelligence, Advanced Control of Auto. System, Data Inference and Applied ML, ML for Signal Proc., Applied Computer Vision, Natural Language Proc., Introduction to Deep Learning, Intro. to Cognitive Robotics, Big Data Science, Microelectronic Devices and Circuits, Digital Circuit Analysis and Design, Nanotechnology and Nanoelectronics

TECHNICAL EXPERIENCE

DOCTORAL RESEARCHER / CONTRASTIVE REPRESENTATION LEARNING Norwegian University of Science and Technology	Sep 2023 — Present Trondheim, Norway
--	--

- Published multiple works on contrastive representation learning in top AI conference with less than 25% acceptance rate
- Contributed a total of 782 hours as a teaching assistant and oral exam sensor across five courses in computer science
- Participated in talks on meta-learning and contrastive representation learning in AI conferences and workshops in Europe

COMPUTER VISION INTERN / SELF-SUPERVISED LEARNING RESEARCH Alteia	Jun 2022 — Aug 2023 Toulouse, France
---	--

- Developed and coded the entire training and evaluation pipeline on 3 different pretext tasks from scratch
- Improved the overall accuracy of a vanilla supervised model by 6.5% after pretraining via self-supervised learning
- Visualized the model's results and evaluation metrics on the WandB platform by refactoring the entire code

GRADUATE TEACHING ASSISTANT / ADVANCED COMPUTER VISION Carnegie Mellon University	Aug 2021 — Dec 2021 Kigali, Rwanda
---	--

- Reduced assignment grading time by 40% through writing scripts and python codes to grade individual submissions
- Facilitated weekly tutorials for 15 graduate students on homework and difficult concepts covered in the classroom
- Graded 50% of the assignments and computed the class distributions for all 15 students in the fall 2021 semester

SOFTWARE DEVELOPER / SATELLITE DATA VISUALIZATION WEB-DASHBOARD Technoserve	May 2021 — Nov 2021 Cotonou, Benin
---	--

- Developed and deployed a web dashboard for satellite data visualization of cashew farms using Python
- Integrated ground data statistics of 1900 plantations in Benin republic with the shapefiles on the web dashboard
- Enhanced user experience by implementing on-click statistics on all 51 communes in the Benin republic

TECHNICAL SKILLS

Tools and Languages	Python, Git, L ^A T _E X, C++, Java, OpenGL, Heroku, JavaScript, Flask, Django, Linux, Docker, MySQL, GLSL, Rust, CuDA, ROS
Data Visualization	Jupyter, Numpy, Matplotlib, Pandas, OpenCV, Blender, Google Earth, Folium, Scikit-learn
Machine Learning	Gaussian Mixture Models, Hidden Markov Models, K-Means, Random Forest, PCA, NLTK
Deep Learning	Multilayer Perceptron, Convolution Neural Network, Recurrent & GAN Network, Pytorch

ACTIVITIES AND ACHIEVEMENT

Awarded Best Poster at NorwAI Innovate Conference, Trondheim Norway	Sept. 2024
Won the Erasmus Mundus Joint Masters Scholarship, France and Norway	June 2021
Best Lagos state candidate in the National Information Technology Development scholarship, Nigeria	Oct. 2020
Awarded the Carnegie Institute of Technology Dean's Fellowship, Pittsburgh, USA	June 2020